Due date: Wednesday 8:50am, Oct. 21, 2026

Any function, library and API in OpenCV is not allowed to use, unless it is mentioned in the question.

1. Histogram (C/C++) (35%)

   Given two grayscale old photos, *Liuchuan800x500.raw* and *Planck403x596.raw*, enhance their contrast using the following methods and **draw the histogram** before and after.

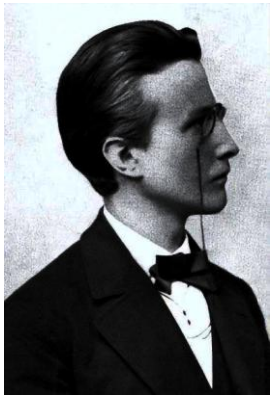
   (You must write your own histogram code, but you can plot it with OpenCV.)

   (1) Apply **Histogram Equalization** to both images. (Figure, 10%)

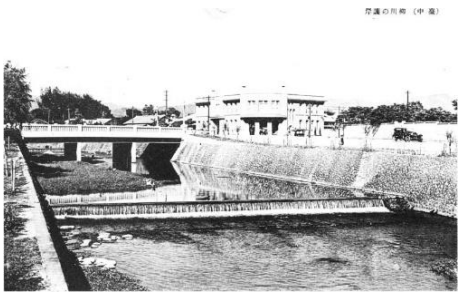
   (2) Apply **Local Histogram Equalization** with different window sizes: 5x5 and 13x13 to both images. (Figure, 10%)

   (3) Apply **Histogram Match** to *Planck403x596.raw*, using *Liuchuan800x500.raw* as the reference image. (Figure, 10%)

   (4) Compare the results of the three enhancement methods. Discuss their differences and explain which method provides the best restoration result and why. (Discussion, 5%)



Planck403x596.raw



Liuchuan800x500.raw

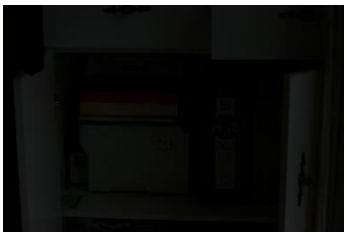
2. Gray-Level Transformations (C/C++) (20%)

   Given two grayscale photos, *dark600x400.raw* and *light512x341.raw*. Perform the following methods on the image and **draw the histogram** before and after. Show, compare, and discuss the outcomes.

   (You must write your own histogram code, but you can plot it with OpenCV.)

   (1) **Nagtive** gray-level transformation (Figure, 5%, Discussion, 5%)

   (2) **Power-law** transformation. Use at least one values less than 1 and one values greater than 1 for the gamma parameters (γ). (Figure, 5%, Discussion, 5%)



dark600x400.raw



light512x341.raw

3. Quantization (C/C++) (15%)

Compress *lena256.raw* and *baboon512.raw* using the following two bit-plane selection method and **draw the histogram** before and after.

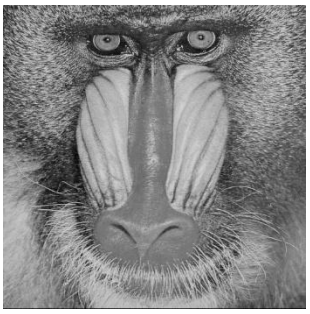
(1) Keep only the top k MSB bit-planes. Find the minimum k to compress image with **PSNR ≥ 35dB**. Show the image and report the selected bit-planes, PSNR.

(2) Display all eight bit-planes and select the bit-planes that contain meaningful image structures. Discard the bit-planes that appear mainly noise-like, and reconstruct the image using the selected bit-planes. Show the image and report the selected bit-planes.

Discuss the differences between the two methods and which one is better. (Figure, 10%; Discussion, 5%)



lena256.raw



baboon512.raw

4. Watermark (C/C++) (30%)

Implement the following two methods to hide your student ID in *baboon512.raw*

(1) Encode your student ID using 8-bit ASCII codes and repeatedly embed the ID (e.g., 123456789123456789...) into the image until all available LSBs are used. Save the encoded image as *baboon512_{your student ID}.raw*. (Figure, 10%)

Note: Encoding must start from the MSB of the character. 1 character=8 LSB of pixel.

(2) Repeat the same student ID embedding task in (1) using **Block Median Quantization Index Modulation (QIM)**. Divide *baboon512.raw* into 8x8 blocks and embed one bit in each block. For each block, calculate its median gray level m, then compute the

quantization index q=m/Δ, where Δ is the **quantization step size** and Δ=8. Use an even q to represent bit 0 and an odd q to represent bit 1. If the parity of q does not match the bit to be embedded, find q+1 or q-1, convert them back to gray levels by multiplying by Δ, and choose the one closest to the original median m. Then shift all pixel values in the block by the same amount so that the block median reaches the selected quantized gray level. Save the encoded image as *baboon512_QIM_{your student ID}.raw*. (Figure, 10%)

(3) Use 6-bit quantization to degrade the **two encoded images**. Then, decode the student ID from the images before and after distortion, compare the contents, and discuss the two watermarking methods and different Δ in (2). (result, 5%, Discussion, 5%)